

KHULNA UNIVERSITY OF ENGINEERING AND TECHNOLOGY

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VLSI

Clocked-load Dynamic Flip Flop

Clocked-load Dynamic FF Features:

- To further reductions to the area and power required for a dynamic FF arise from the use of a ratioless inverter with a clocked-load.
- The circuit comprises a pass transistor and an inverter.
- Low area due to the use of enhancement type transistor. But it reduces the output voltage
- The power reduction arises because power is only consumed when clock is high; it is only during this time that the inverter load , T3, turns on, creating a current path between 5 V and 0 V.
- Vout(high) = Vp – Vte = 5 – 1 = 4 V But due to body effect, Vout = 3.12 V and the 2nd output, Vout = 1.9 V due to 3.12 as input.

Clocked-load Dynamic Flip Flop

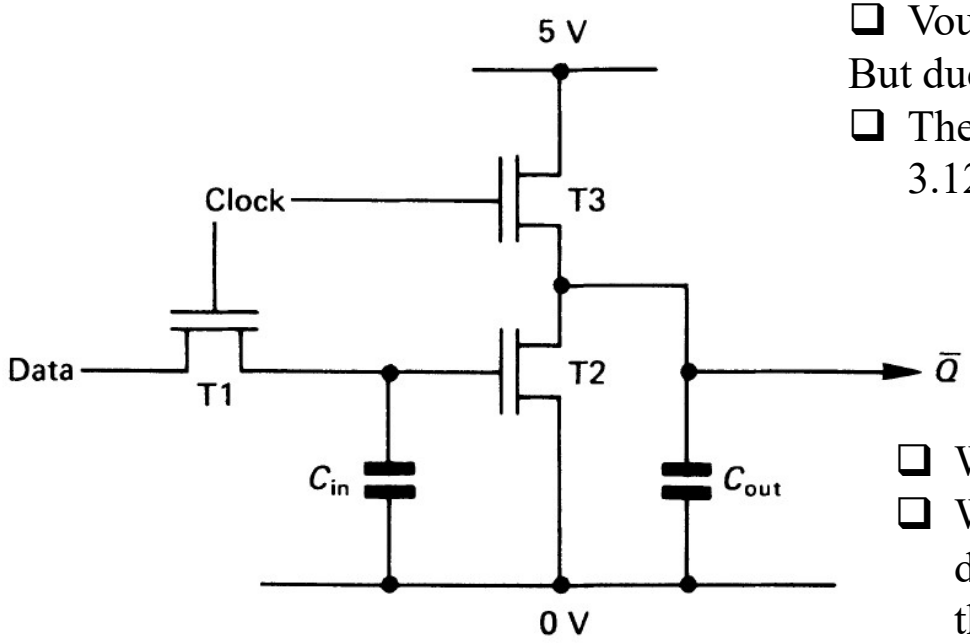


Figure 4.19 A clocked-load dynamic flip flop

- Vout(high) = Vp – Vte = 5 – 1 = 4 V

But due to Body effect the Vout = 3.12 V (app)

- The 2nd output, Vout = 1.9 V approximately due to 3.12 as input.

- When clock = 0 V then T1, T3 off
- When clock high then T1, T3 high then deepens on the data input, if data is low then low voltage passes to the gate of T2 then T2 off and Vout = 5 – Vte V, clock off 5 – Vte voltage is maintained.
- When both T2 and T3 are ON, the output voltage cannot be determined correctly.

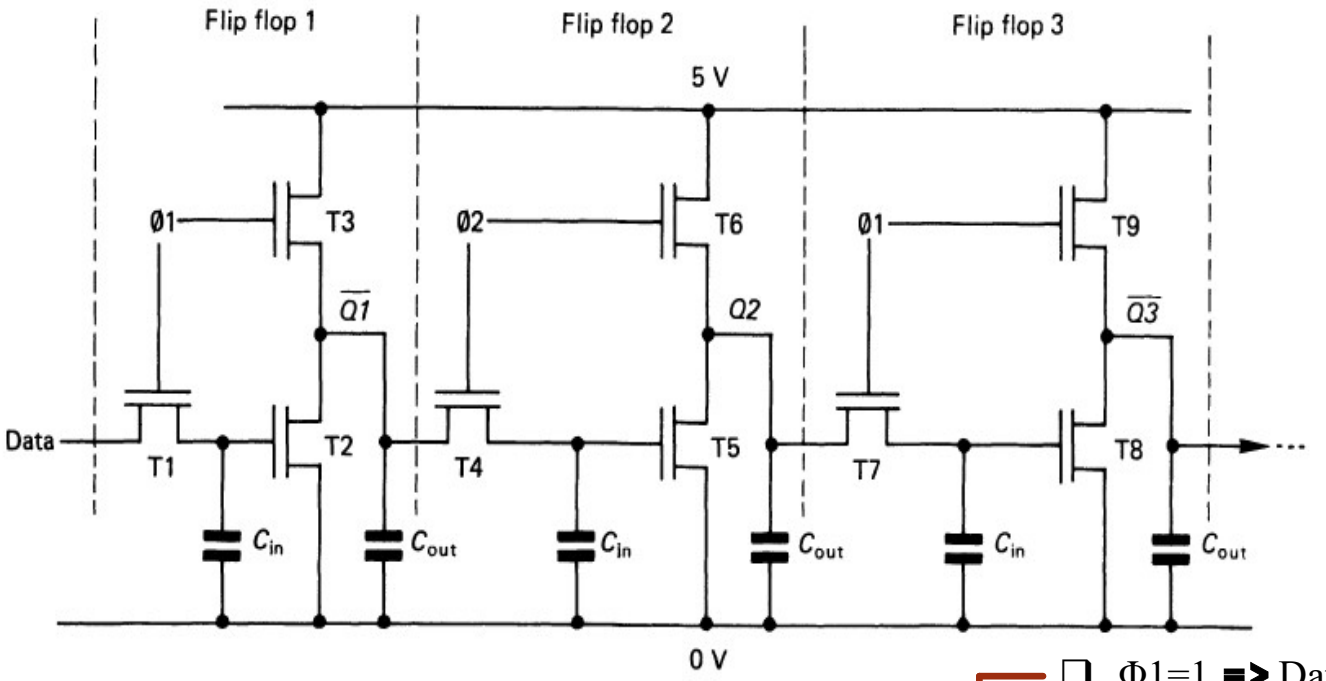


Figure 4.20 A clocked-load shift chain

- ❑ Φ1=1 => Data=0 => $\overline{Q1}$= 5-Vte. /// Φ2=0
- ❑ Φ1=1 => Data=high => $\overline{Q1}$= Indt…. /// Φ2=0
- ❑ Φ1=0 => Data=high => $\overline{Q1}$= 0 through discharge…. /// Φ2=0
- ❑ Φ1=0 => Data=high => $\overline{Q1}$= 0 through discharge…. /// Φ2=1 => Cin= 0 through discharge(T4-T2)
- ❑ Φ1=0 => Data=0 => $\overline{Q1}$= 5-Vte …. /// Φ2=1 => Cin= Charge share (Cout & Cin) => Low V

Here, the successful transfer of a high level to the gate of T5 is critically dependent upon the relative magnitude of C_{out} and C_{in}. Before T4 is turned on, the charge Q_{out} stored on C_{out} at $\overline{Q1}$ is

$$Q_{out} = C_{out}(V_p - V_{te})$$

When T4 turns on, this charge Q_{out} is shared between the two capacitors which are in parallel, and the new voltage level V_{out} across both capacitors is

$$V_{out} = \frac{Q_{out}}{C_{in} + C_{out}} = \frac{C_{out}(V_p - V_{te})}{C_{in} + C_{out}}$$

In fact, V_{out} has to significantly exceed V_{te} to turn on enough current in T5 (when $\emptyset 2$ is removed) to avoid incurring a long discharge time. Assuming $V_p - V_{te}$ is approximately 3 V and a minimum V_{out} of 2 V after the charge transfer, then $C_{out} > 2C_{in}$.

RAM-Random Access Memory

Important Features of RAM:

- A group of registers where only one register is accessed at a time and where the time obtain or alter information in any register is similar is referred to as a RAM.
- Access time is similar for all register
- Each register has its own identity
- 2^n line store requires n-bit address line
- 2^n n-input nand gates are required to store one bit
- It can access read/write operation
- Address decoder is required to select a line
- The address is split into two parts usually referred to as the X and Y bits, and each part is separately decoded. Although this reduces the decoding logic to 2^X X-input plus 2^Y Y-input nand gates, where X + Y = n, there is only a total of $2^X + 2^Y$ select signals, thus total no. Of nand gates are reduced.

  2^X row matrix and 2^Y column matrix.

RAM: General schematic of a 2^n by one bit store

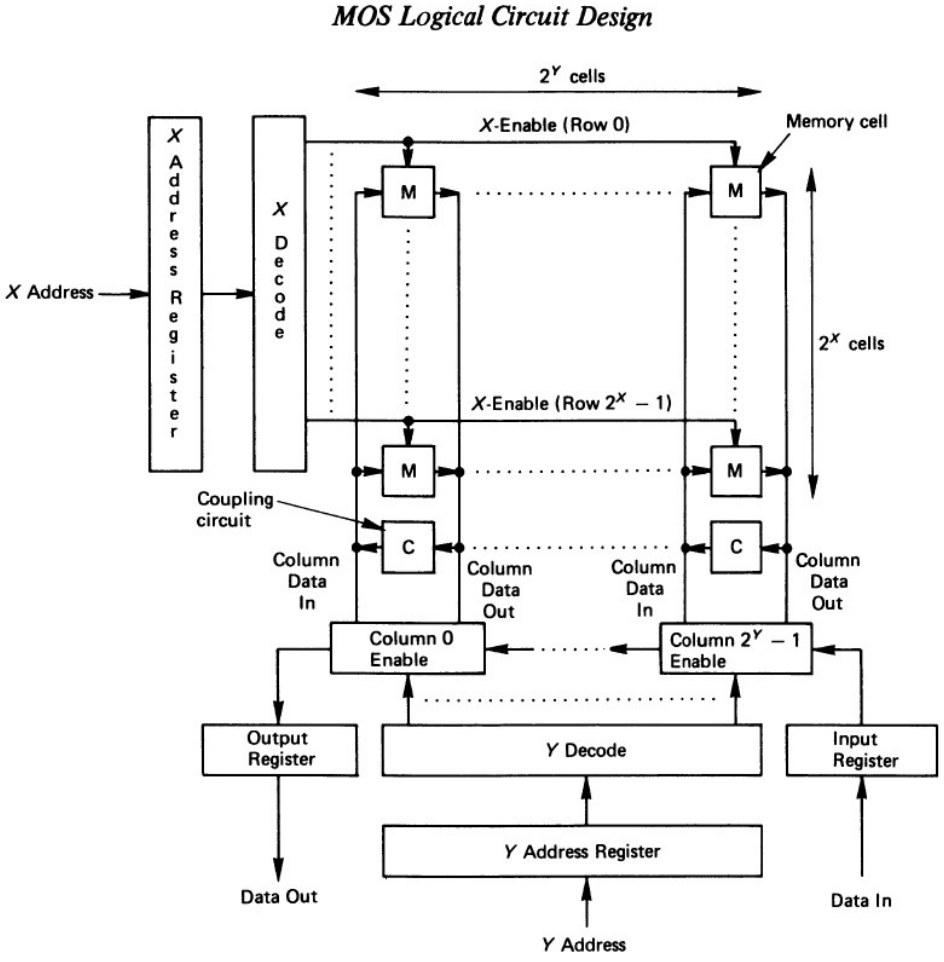


Figure 4.21 General schematic of a 2^n by one bit store

A large number of signals are required to drive a store of significant size and it is often necessary to reduce them. Most often this reduction is affected by using the same lines for the *X* and *Y* address and by the use of common (bidirectional) data input/ output lines. Sharing the address lines relies on the fact that arow of cells is first selected by the *X* decoder and then one of these cells is enabled for reading from or writing to by the *Y* decoder. Thus, the *X* address is presented first and clocked into the *X* decode register. This allows the row decode and selection to operate while the *Y* address is presented and clocked into the *Y* decode register.

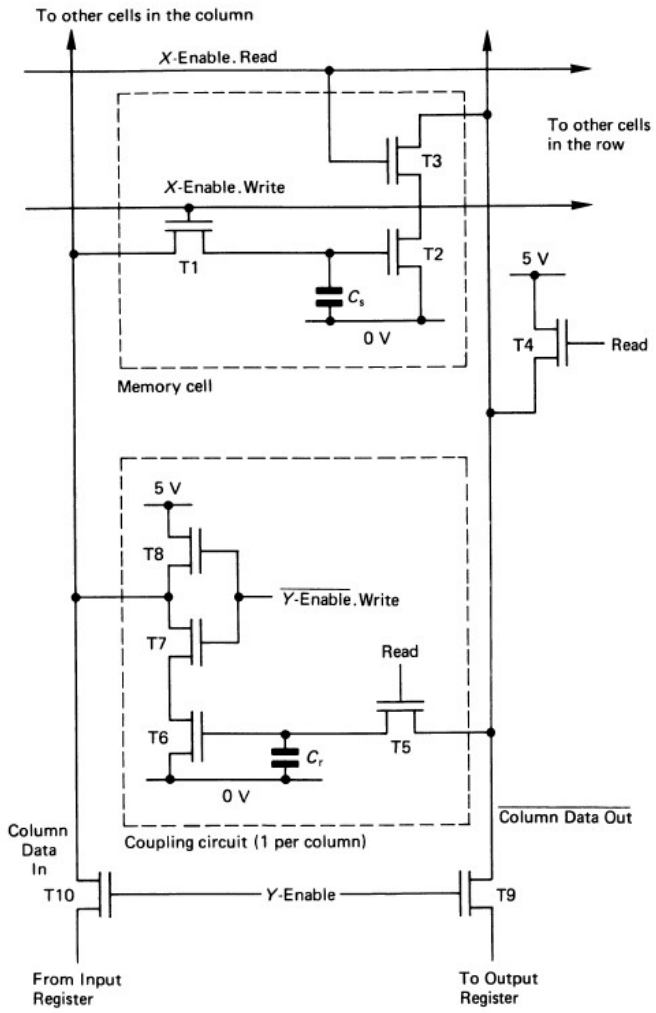


Figure 4.22 A three-transistor per bit memory cell and its control signals